



OPEN Rapid antimicrobial susceptibility testing using carbon screen printed electrodes in a microfluidic device

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The development of rapid, sensitive, and affordable antimicrobial susceptibility testing (AST) is essential for controlling antibiotic overuse, thereby creating a critical checkpoint for the emerging antimicrobial resistance threat. Here, we introduce a novel method of electrochemical monitoring of bacterial growth in a diluted low-conductivity nutrient medium for rapid susceptibility testing using impedance spectroscopy. The method works on the change in charge transfer resistance exhibited by bacteria in response to antibiotics. The proposed Electrochemical Microfluidic device (ϵ - μ D) employs low-cost carbon screen-printed electrodes and uses a simple microfluidic geometry. We explored the utilisation of a diluted nutrient medium as an electrolyte since it provides a higher charge transfer baseline signal for better sensitivity and supports the growth of the bacteria required for detection. The method enables sensitive detection of bacteria even at a low density of 84/mm² in three hours of incubation time. For proof of concept, bacteria such as *Escherichia coli* and *Bacillus subtilis* were used, and the efficacy of the ampicillin and tetracycline drugs were tested. The experiments were done with the spiked urine samples, which correlated well with the controlled sample. The proposed system enhances the accessibility and affordability of rapid susceptibility testing, enabling its widespread use.

Keywords Impedance spectroscopy, Antimicrobial susceptibility testing, Microfluidics, Carbon electrodes, Charge transfer resistance

Antimicrobial resistance (AMR), a defensive technique of microorganisms against antimicrobials, has been identified as one of the top ten threats to global health by the World Health Organization (WHO)¹. It is estimated that 4.95 million deaths globally (2019) have been associated with bacterial AMR, with its highest incidence in limited-resource regions². Antimicrobial susceptibility testing (AST) determines the resistance and susceptibility profile of available drugs for specific infections and ensures their effectiveness in treatment³. AST is an important strategy in mitigating the burden of AMR by controlling the irrational use of antibiotics^{4,5}.

The conventional gold-standard culture-based AST is labour-intensive and has a longer turn-around time (48–72 h)⁶. Most bacterial infections spread faster, so timely detection of infection and administration of appropriate drugs is essential for better patient outcomes. Several genotypic methods based on polymerase chain reaction principles are emerging to identify resistant genes in the bacteria. Although these methods produce rapid results with high throughput, they have several challenges: the requirement of sophisticated instruments, continuous updation of resistance profiles, and confirmation of results from phenotypic testing^{7,8}. Other techniques have been developed for rapid AST, like determining bacterial growth by microscopy or spectroscopy, biochemical profiling by mass spectrometry, and detecting micromotions of bacteria using a cantilever in Atomic force microscopy. However, these techniques are cost-intensive and require highly skilled technician, which makes them challenging to implement in resource-constrained places^{8–10}.

Electrochemical impedance spectroscopy is a fast, robust, and label-free diagnostic tool for bacterial detection and AST¹¹. Sensors using inter-digitated electrodes (IDEs) have widely been reported to enable rapid bacterial detection and AST where sample volume is reduced to a few μ L^{12–14}. Using a different approach, Hannah et al., utilised commercially available gold screen-printed electrodes with an agarose gel deposit for rapid AST detection¹⁵. The major challenges of these techniques are the device fabrication and their cost since they use precious metals for fabricating electrodes. The incorporation of electrochemical detection in microfluidic systems has gained attention, expediting detection using small sample volumes, ranging from a few nanoliters

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to picoliters^{5,16}. These electrochemical microfluidic devices focus on capturing a single to a few tens of bacteria, giving high sensitivity^{17–20}.

In this work, we propose a rapid and affordable phenotypic AST using electrochemical impedance spectroscopy (EIS) in screen-printed carbon electrodes and simple microfluidic geometry. We termed our device as ϵ - μ D (electrochemical microfluidic device). Further, we have used a dilute low-conductivity nutrient medium as an electrolyte, which provides a higher impedance baseline signal while supporting the growth required for the sensitive determination of susceptibility. We show that the charge transfer resistance decreases with respect to time due to bacterial growth, even in the presence of a low number of bacteria (#84/mm²). As the ϵ - μ D depends on bacterial growth, the effect of cell growth with antibiotics was correlated with the change in impedance of the electrical signal. We establish the proof-of-concept of the proposed device using gram-negative (*E. coli*, ATCC 25922) and gram-positive bacteria (*B. subtilis*, ATCC 6051) against ampicillin (bactericidal) and tetracycline (bacteriostatic) antibiotics. We also tested the efficacy of the device to determine the AST of spiked *E. coli* from the urine sample matrix against tetracycline. With its high sensitivity and low-cost fabrication, the proposed device has the potential to democratise rapid susceptibility testing. The fabricated device is fast, affordable, feasible, sensitive, and easy to handle, and it meets all the criteria prioritised by the WHO for rapid AST diagnosis²¹.

Results and discussion

Optimising electrolyte with bulk solution

The selection of appropriate electrolytes is a crucial step in developing EIS sensors. In bacterial detection, low-conductivity materials are preferred as electrolyte since they provide a higher sensitivity of the sensor. These materials allow higher absolute impedance values and mainly exhibit enhanced sensitivity to impedance changes compared to high-conductivity medium/buffers²². We aim to optimise the electrolyte composition to provide a higher impedance baseline signal while supporting bacterial proliferation necessary for AST sensing. To achieve this, the impedance and growth profile for the different strengths (10, 25, 50, 75, 100% (v/v) of medium/water) each of Luria-Bertani (LB) and tryptone nutrient medium (TNM) were investigated. Platinum was used as the working and counter electrode in a 10 mL sample volume (Supplementary Fig. S1). The impedance value of 10% TNM was strikingly higher (~3 times) than 10% LB (Supplementary Fig. S2a). This confirmed that the TNM has a higher baseline signal, and further experiments were conducted with TNM. Figure 1a shows the Nyquist plot for the different strengths of TNM. The impedance value increased as the tryptone strength decreased. The conductivity measurement also supports the impedance results (Supplementary Table S1). The effect of diluted tryptone medium on bacterial growth (*E. coli*) was calculated using an exponential fit to the log phase of growth curves. It was found that the doubling time was higher in the 10% TNM (88 ± 1 min) than in the relatively rich medium, i.e. 100% TNM (52 ± 3 min) (Fig. 1b). A relatively low turbidity was observed in the 10% TNM (Supplementary Fig. S3a). However, 10% TNM could support growth for high bacterial concentration (~10⁷ CFU/mL), enabling AST determination (Supplementary Fig. S3b). Though 5% TNM showed the highest impedance (Supplementary Fig. S2b), we found that it does not support bacterial growth, and there is no turbidity for 5% TNM (Supplementary Fig. S3b and a). So, 10% TNM is an excellent choice of electrolyte because it shows a higher impedance signal (~10 times than absolute tryptone medium) and is suited for bacterial growth. Therefore, in all the experiments, this was used as an electrolyte.

The experimental data from the Nyquist plot was fitted to the equivalent circuit model (ECM) to extract specific analytical parameters. The classical Randle's circuit model was typically used to obtain parameters such as solution resistance (R_s) at high frequency, capacitive double layer C_{DL} / charge transfer resistance R_{ct} at low frequency, and diffusional Warburg impedance (Z_w) at very low frequency²³. R_s measures a relatively large change in ionic components of the bulk solution, and R_{ct} measures resistance at the electrode-electrolyte interface. In this study, an improved Randle's circuit model was used to fit 10% TNM, as shown in Fig. 1c. The improved ECM has two key modifications: (a) Replacement of C_{DL} with constant phase element Q_{DL} . (b) Replacement of Z_w with constant phase element Q_D to capture diffusion process. These modifications were made based on the experimental condition, mainly to capture the non-ideality of 10% TNM solution. Long et al., have replaced double layer capacitance with constant phase element (CPE) to incorporate the roughness and porous electrode²⁴. ECM was found to fit the experimental data clearly well as represented by solid line in Fig. 1d (blue line).

Next, the effect of bacterial presence in the impedance spectrum of electrolyte (10% TNM) was investigated. Different concentrations of bacteria (10⁸, 5 × 10⁸, 10⁹ CFU/mL) were added to the electrolyte, and the impedance spectrum was analysed immediately (Fig. 1d). The impedance values decreased as the bacterial concentration increased. Bacterial cells at high concentrations facilitate the transfer of electrons between working and counter electrodes, acting like pearl chains²⁴. The parameters obtained by fitting the ECM to the experimental data are given in Supplementary Table S2. The improved Randle's circuit fit the experimental data well ($\chi^2 < 0.05$). Further, as detailed in the upcoming section, the normalised impedance signal calculated from the extracted R_{ct} was used as a response signal for bacterial detection and susceptibility testing.

Electrochemical microfluidic device (ϵ - μ D) design and bacterial detection

Figure 2a shows a schematic of the ϵ - μ D employed for bacterial detection and AST (Supplementary Fig. S4a represents the setup). The parameters such as the nature of the electrode, its area, the distance between the working electrode (WE) and counter electrode (CE), and sample volume greatly influence the impedance signal. Here, we have used low-cost carbon electrodes suitable for high-throughput screening. Reducing the electrode area and sample volume and increasing the gap between electrodes can result in a higher impedance baseline, giving better sensitivity. However, it is essential to strike a balance in choosing these factors optimally to obtain a reliable signal from the potentiostat, especially when working with low-conductivity electrolytes. The optimal

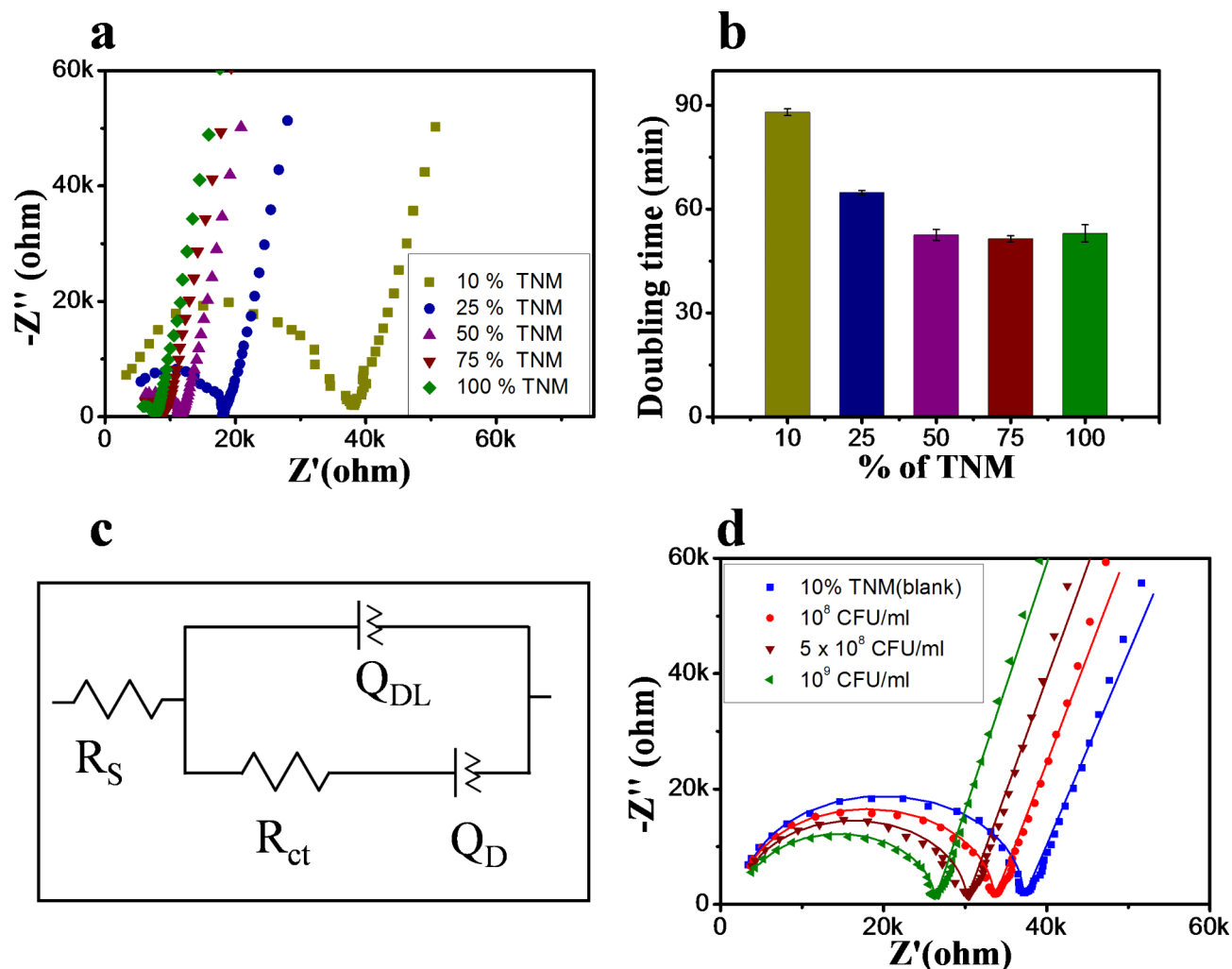


Fig. 1. Optimization of the electrolyte using bulk solution (10 mL) (a) Nyquist plot showing the effect of different tryptone nutrient medium (TNM) strengths on the impedance signal. (b) Effect of varying strength of TNM on the bacterial growth, showing the compatibility of bacterial growth on 10% TNM (c) Equivalent circuit model developed to obtain the charge transfer resistance signal from the Nyquist plot (d) Nyquist plot with the variable concentration of bacteria in 10% TNM with solid line representing the model fit.

dimensions of the microfluidic device were (1.5 mm (W) $\times$ 14 mm (L) $\times$ 0.5 mm (D)) with an electrode width of 0.5 mm. The characterization of the PDMS microfluidic channel using profilometry and microscope showed that the dimensions were 1.5 ± 0.015 mm (W) $\times$ 14 ± 0.05 mm (L) $\times$ 0.500 ± 0.001 mm (D). The height and width of the electrode was 531 ± 0.64 μ m and 4.7 ± 0.12 μ m. The device used two sets of electrode pairs, WE1-CE1 and WE2-CE2, which served as duplicates (Fig. 2a). The electrodes were functionalised with poly-L-lysine (PLL). This cationic polymer imparts a positive charge to the electrodes, thereby effectively immobilising bacteria by electrostatic interaction on the surface of the electrodes^{25,26}. The sample containing bacteria was introduced into the ϵ - μ D, incubated for 30 min to immobilise them, washed with electrolyte, ultra-low conductivity medium (Supplementary Table S1, 10% TNM – 560 μ S/cm) to remove the sample matrix, and the impedance was measured over time (details in the method section) (Fig. 2b). The change in the impedance was correlated with the bacterial concentration.

The use of low conductive growth medium/buffer has gained more popularity in bacterial detection and susceptibility testing, where the changes in solution resistance have been correlated with the efficacy of antibiotics^{11,27}. On the other hand, some studies have focused on utilising the low conductivity PBS buffer/deionised water to directly detect bacteria with a change in charge transfer resistance as a response signal^{24,28}. While this method can yield more sensitive results due to lower conductivity, it cannot support bacterial growth. In our novel approach, we use a growth-supporting medium to observe charge transfer resistance as a response signal. This is achieved by diluting a low-conductivity nutrient medium to obtain an ultra-low conductivity medium (10% TNM – 560 μ S/cm) where charge transfer resistance is dominant. Moreover, 10% diluted TNM could support bacterial growth as well. This provides an ideal medium for obtaining higher sensitivity than high strength-low conductivity medium (100%) and non-growth supporting buffer/DI.

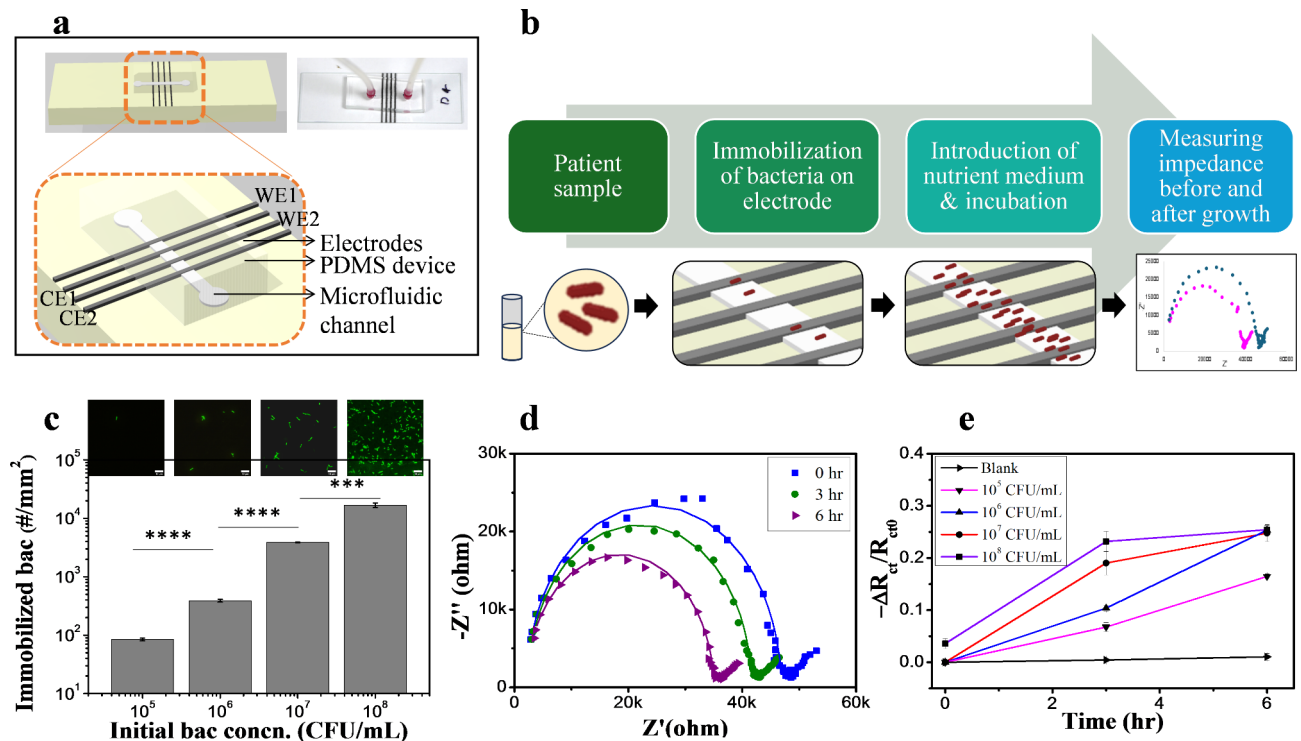


Fig. 2. Bacterial detection using the proposed electrochemical microfluidic device (ϵ - μ D). (a) Schematic of the electrochemical microfluidic device employed for bacterial detection and susceptibility testing, WE- working electrodes; CE- counter electrodes. (b) Schematic illustrating the working principle of ϵ - μ D. (c) Determination of immobilised bacteria (after washing) as a function of initial bacterial concentration. Inset figure displays representative fluorescent microscopic images for each concentration using a 50X objective. The scale bar is 10 μ m. **** denotes $p < 0.0001$ and *** denotes $p < 0.001$ (d) Nyquist plot showing the effect of bacterial growth over time on the ϵ - μ D with an initial bacterial concentration of 10^6 CFU/mL (e) Normalized impedance signal ($-\Delta R_{ct}/R_{ct0}$) over incubation times (3, 6 h) for different initial bacterial concentration (10^5 – 10^8 CFU/mL). Only immobilised bacteria are responsible for the observed NIS. Even it could detect #84/ mm^2 immobilised bacteria after washing, initially corresponding to 10^5 CFU/mL.

The number of bacteria immobilised on the device was measured for different initial concentrations of bacteria using fluorescence microscopy (Fig. 2c). This number represents the actual bacteria that are responsible for the impedance change caused by bacterial growth. Insets in Fig. 2c show microscopic images of immobilised bacteria for different initial concentrations. The number of immobilised bacteria was 84 ± 6 per mm^2 for an initial concentration of 10^5 CFU/mL (1,764 immobilised bacteria/device where device area is 21 mm^2). This increases logarithmically with the concentration.

The electrical response to bacterial growth on the ϵ - μ D over time was monitored. For this, impedance measurement was carried out immediately after introducing electrolyte corresponding to $t = 0$ h. The device was then incubated for specified intervals ($t = 3$ and 6 h), and the spectrum was measured. We experimented with different initial concentrations of bacteria (10^5 – 10^8 CFU/mL) and measured EIS over the incubation period of 6 h. Figure 2d shows a representative Nyquist plot at different incubation times for a 10^6 CFU/mL initial concentration. Similar plots for other initial bacterial concentrations are shown in Supplementary Fig. S5. This plot reveals that bacterial growth causes a significant change in the impedance value. This is evident by tracking the elbow point, connecting the semi-circle and linear tail. With increased incubation time, the elbow point shifts from right to left, signifying the decrease in charge transfer resistance. Bacterial growth leads to the production of ionic metabolites, which strikingly alter the impedance characteristic of the electrochemical cell^{11,18}. The observed change in charge transfer resistance is attributed to the release of metabolites during bacterial growth and also the bacterial presence itself acting as a pearl chain²⁴ (Fig. 1d). The solid lines in Fig. 2d represent the proposed model fit to the experimental data as explained in ‘Optimizing electrolyte with bulk solution’ section. The improved model developed in the bulk solution experiments was found to fit well with the experimental data of the ϵ - μ D and is used to extract R_{ct} . The normalised impedance signal (NIS), based on the change in charge transfer resistance, was used as a response signal, as defined in the materials and method section.

Figure 2e shows the NIS ($-\Delta R_{ct}/R_{ct0}$) for different initial concentrations of bacteria (10^5 – 10^8 CFU/mL) over the incubation period of 6 h. Blank experiments (without bacteria) were performed to ensure the observed charge transfer resistance was only due to bacterial growth and not to artefacts like evaporation losses. No changes in charge transfer resistance for blank experiments (0.010 ± 0.006) were observed during the incubation times till 6 h, thereby confirming that only bacterial growth determines NIS. A distinct NIS value was obtained for the test

samples with bacteria. NIS at $t = 3$ h increased with initial bacterial concentration. A NIS value of 0.067 ± 0.008 was observed even at $\#84/\text{mm}^2$ (initial concentration: 10^5 CFU/mL), which was found to be significantly higher than the blank experiment ($n = 3$, $p < 0.001$). This confirms the high sensitivity of the proposed device. The NIS value increases with incubation time $t = 6$ h for 10^5 and 10^6 CFU/mL. For higher bacterial concentrations (10^7 and 10^8 CFU/mL), the NIS value saturates at $t = 3$ h. Figure 2e serves as a calibration chart to quantitatively estimate bacteria. However, it is important to note that the quantitative estimation reported here is bacterial-specific.

We calculated the limit of detection (LOD) using the calibration curve of the normalized impedance signal at $t = 3$ h with immobilized bacteria (Supplementary Fig. S6). The LOD was estimated to be 525 CFU ($3.3 \sigma/S$), where σ represents the standard deviation of the response and S is the slope of the linear regression equation. The initial bacterial concentration corresponding to the LOD (525 CFU) was determined to be 2×10^4 CFU/mL, as calculated using the linear equation obtained from Fig. 2c.

Antimicrobial susceptibility testing using the proposed ϵ - μ D

After establishing that the proposed ϵ - μ D can detect the bacteria successfully, its utility for AST was investigated. Two identical microfluidic devices were employed to achieve the susceptibility testing, and the steps involved in the bacterial detection described in the previous section (Fig. 2b) were followed. The detailed steps are given as a flowchart in Supplementary Fig. S7. In the first device, the experiment was performed without antibiotics, referred to as a positive control (PC). The impedance change obtained using a PC device is termed as ΔZ_1 . In the second, the antibiotic was added to the 10% TNM and introduced into this device, termed AST-device (Antibiotic susceptibility testing device). The change in impedance observed before and after incubation with AST-device is termed ΔZ_2 . A PC device was used to confirm the bacterial presence and metabolite production, and it served as a control for each test experiment. It is anticipated that if cells are susceptible, cell division and metabolite production will not occur in response to drug action. Hence, the impedance change (ΔZ_2) will be low for susceptible bacteria and $\Delta Z_1 > \Delta Z_2$. On the contrary, if the cells are resistant, cell division and metabolite production progress as usual, and we would have $\Delta Z_2 \approx \Delta Z_1 > 0$.

To determine susceptibility, experiments were performed using strains of *E. coli* susceptible and resistant to ampicillin. An initial bacterial concentration of 10^7 CFU/mL was employed in all susceptibility experiments. Figure 3a and b shows the Nyquist plot of susceptible and resistant *E. coli* at ($t = 0$ h) and after incubation ($t = 3$ h). It is evident that the elbow point, which is a measure of the R_{ct} , remains constant after incubation with $10 \mu\text{g}/\text{mL}$ of ampicillin (Fig. 3a). In contrast, for resistant bacteria exposed to $50 \mu\text{g}/\text{mL}$ ampicillin, the elbow point

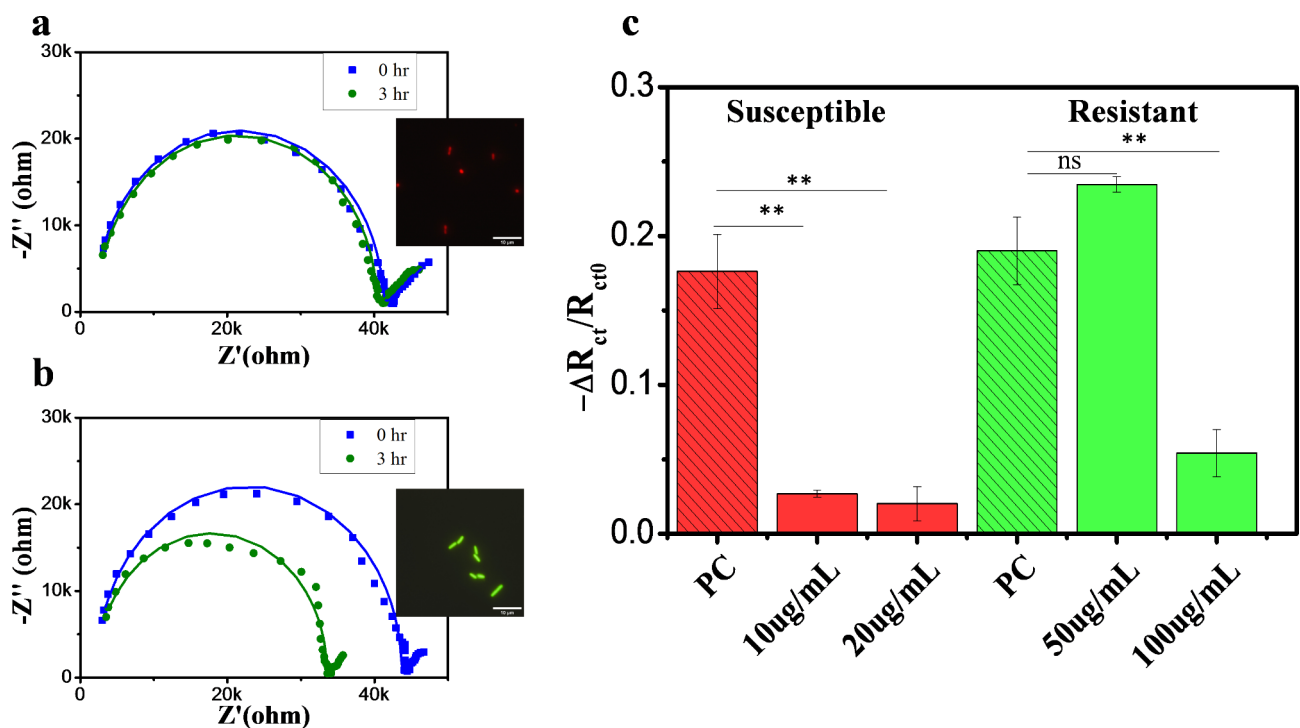


Fig. 3. Antimicrobial susceptibility testing using proposed ϵ - μ D (a) Nyquist plot before and after an incubation time of 3 h for susceptible bacteria at $10 \mu\text{g}/\text{mL}$ ampicillin. Inset image: microscopic observation at 100X objective showing red fluorescence, indicating the dead bacteria on the device. Scale bar: $10 \mu\text{m}$ (b) Nyquist plot over incubation time for resistant bacteria at $50 \mu\text{g}/\text{mL}$ ampicillin. Microscopic observation showing green fluorescence, indicating the viable bacteria on the device displayed in the inset (c) NIS for susceptible and resistant bacteria with different concentrations of ampicillin. Here, PC denotes positive control, bacteria without antibiotics. ** indicates $p < 0.01$, ns indicates not significant ($p > 0.05$).

moves to the left (Fig. 3b). Decrease of R_{ct} denotes that the bacteria are growing even at this high concentration of antibiotic, which confirms that they are resistant to ampicillin. To test the viability of bacteria, the device was stained with SYTO9/ propidium iodide (PI) dyes (BacLight bacterial viability kit) after the incubation for both cases with or without antibiotics (Inset of Fig. 3b and a). Susceptible bacteria emit red fluorescence (Inset of Fig. 3a), revealing its loss in membrane integrity due to ampicillin action and allowing its penetration to PI dye. Whereas, resistant bacteria emit green fluorescence (Inset of Fig. 3b), indicating cell integrity and non-penetration of cell impermeable PI stain.

The solid lines in Fig. 3a and b show that the model fits the experimental data, and extracted R_{ct} was used to determine the NIS. Susceptibility experiments were performed with different concentrations of ampicillin for both strains till the NIS was found to be low. For susceptible strains, NIS values at 10 and 20 $\mu\text{g/mL}$ were significantly lower from the PC. For resistant bacteria, NIS at 50 $\mu\text{g/mL}$ was not statistically significant from the PC, as shown in Fig. 3c. Only at 100 $\mu\text{g/mL}$, the resistant bacteria showed a low impedance change when compared to PC.

According to the Central Laboratory Standard Institute (CLSI), for *E. coli*, when the minimum inhibitory concentration (MIC) of ampicillin is greater than 32 $\mu\text{g/mL}$, the bacteria is defined as a clinically resistant species. Thus, the observed resistant/susceptible profile was found to align well with the standards set by CLSI²⁹. The susceptibility and resistance profile of the bacteria were validated using a standard disc diffusion test, following CLSI guidelines. Supplementary Figure S8 shows the disc diffusion test for the susceptible and resistant bacteria. A zone of inhibition (ZOI) of 17.4 ± 0.2 mm with ampicillin (10 $\mu\text{g/disc}$) was observed for the susceptible strain (Supplementary Fig. S8a), which is greater than 17 mm, set by CLSI for the susceptible strain. For resistant bacteria, no ZOI was observed (Supplementary Fig. S8b). These results demonstrate that the proposed $\epsilon\text{-}\mu\text{D}$ could successfully determine the resistant/susceptible profile of bacteria and match well with the CLSI standards.

The versatility of the proposed $\epsilon\text{-}\mu\text{D}$ was tested through experiments with tetracycline, a bacteriostatic antibiotic, and with *Bacillus subtilis*, a gram-positive bacteria. Figure 4 shows the NIS of *E. coli* and *B. subtilis* exposed to each of 10 $\mu\text{g/mL}$ of ampicillin and 5 $\mu\text{g/mL}$ of tetracycline. For *E. coli*, the NIS of the device treated with tetracycline and ampicillin was statistically significant from the PC ($n=3$, $p<0.05$), revealing that the

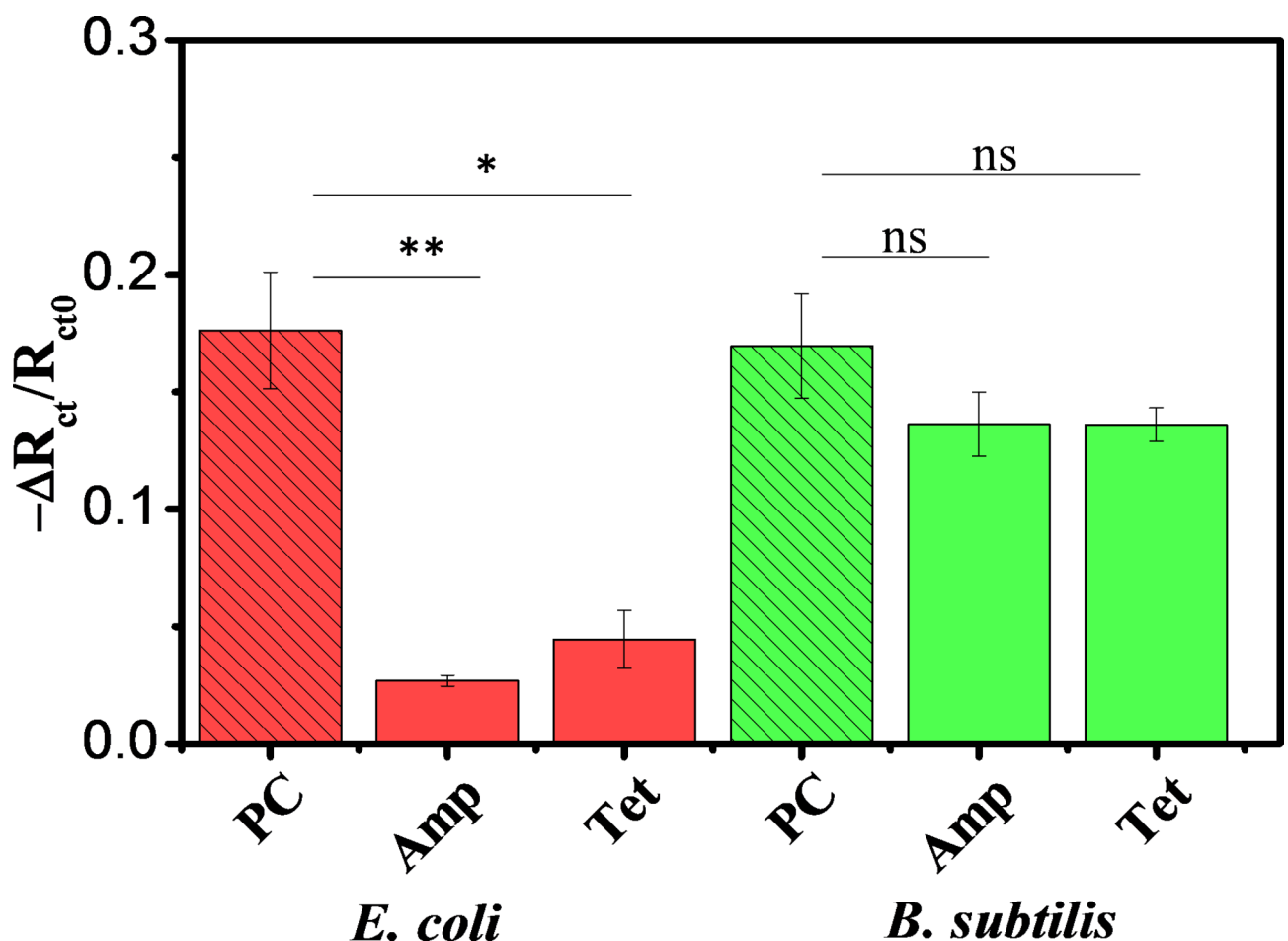


Fig. 4. Antimicrobial susceptibility testing for *E. coli* and *B. subtilis* against ampicillin and tetracycline antibiotics. The NIS plot shows the adaptability of the proposed $\epsilon\text{-}\mu\text{D}$ for other antibiotic and bacterial types. PC: positive control without antibiotic, amp: 10 $\mu\text{g/mL}$ of ampicillin and tet: 5 $\mu\text{g/mL}$ of tetracycline. ** indicates $p<0.01$, * indicates $0.01<p<0.05$, ns: not significant ($p>0.05$).

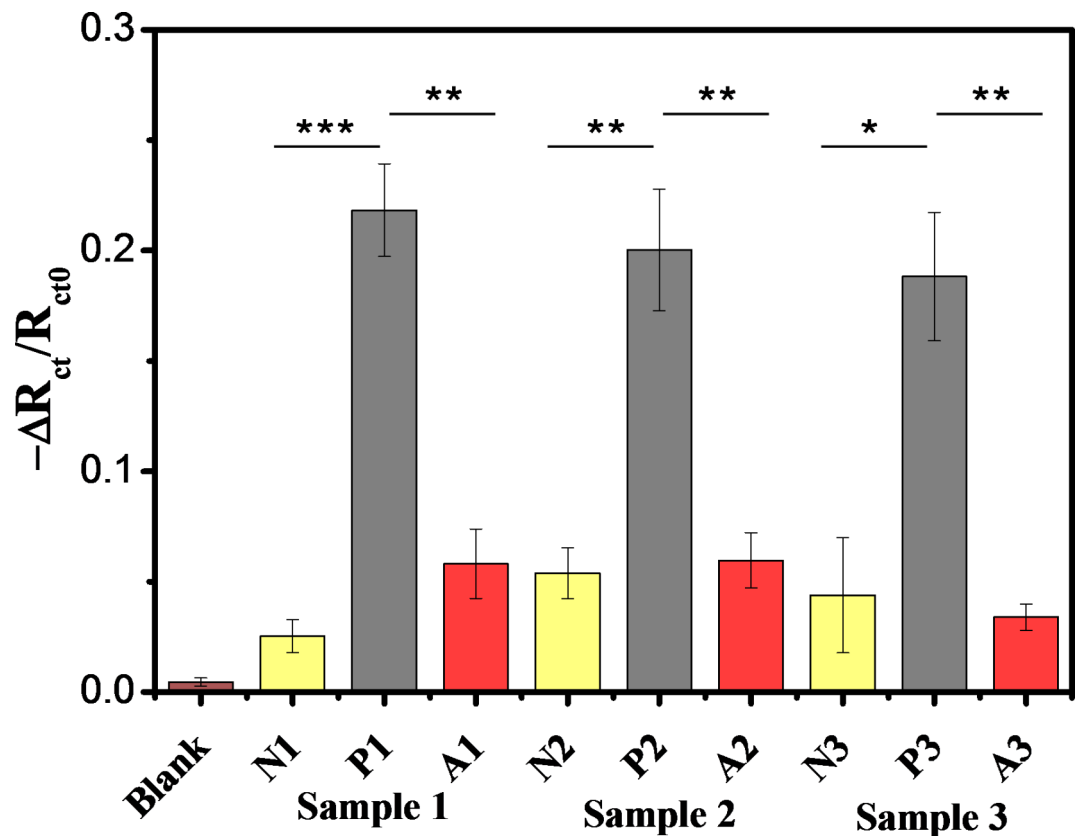


Fig. 5. Normalized impedance signal ($-\Delta R_{ct}/R_{ct0}$) to test the efficacy of the proposed method in human urine samples. Blank represents NIS in 10% TNM. A negative sample (N1, N2, N3) indicates a urine sample without spiking of *E. coli*. The positive sample (P1, P2, P3) denotes 10^7 CFU/mL spiking of *E. coli* in the urine sample. The positive sample is treated with antibiotic tetracycline: 5 μ g/mL, denoted as A1, A2, and A3. *** indicates $p < 0.001$, ** indicates $p < 0.01$, * indicates $0.01 < p < 0.05$.

bacteria were susceptible to both antibiotics. The ZOI was found to be 20 ± 0.4 mm (Supplementary Fig. S8a) with tetracycline (30 μ g/disc), which is in line with CLSI's guideline of $ZOI \geq 15$ mm for susceptible strain²⁹. For *B. subtilis*, NIS for both antibiotics was not statistically significant from the PC, indicating that *B. subtilis* was resistant to both drug types. Hence, the proposed device also worked well for gram-positive species, such as *B. subtilis*. The obtained susceptibility profiles were validated with a disc diffusion test where no ZOI was formed for *B. subtilis* for both antibiotics (Supplementary Fig. S8c). These results reveal that the proposed ϵ - μ D is highly versatile and applicable to both bacterial types (gram-positive and gram-negative bacteria) and drug types (bacteriostatic and bactericidal), as it works on the growth of the bacteria.

Validation using human urine samples

The proposed ϵ - μ D was tested for its ability to detect bacterial presence in human urine samples, followed by AST. The urine samples were collected from the Institute Hospital at the Indian Institute of Technology Madras, India, from both men and women undergoing routine urine examinations. The pathology reports were used to confirm that the samples were bacteria-free. Hence, the samples were spiked with bacteria to perform AST. Three different sets of experiments were conducted: (i) Negative (N) sample: Urine sample without bacteria (ii) Positive (P) sample: urine sample with spiked bacteria (*E. coli*-ATCC25922, 10^7 CFU/mL) (iii) Antibiotic (A) treated sample: AST-device with a positive sample treated with tetracycline (5 μ g/mL). The urine samples, with or without bacterial spiking, were directly loaded into the device without pre-treatment and subjected to bacterial detection and susceptibility testing. Figure 5 shows the NIS obtained from the urine samples. The blank sample without urine showed a negligible NIS value, whereas the negative urine samples (N1, N2, N3) showed a low NIS value. This is due to different ions in the sample matrix. All positive urine samples spiked with bacteria (P1, P2, P3) showed a striking higher value of NIS ($p < 0.05$) when compared to negative samples. A minimum value of NIS was obtained for the antibiotic-treated device (A1, A2, A3) when compared to positive samples ($p < 0.01$), confirming its susceptibility against tetracycline. We conclude that the proposed ϵ - μ D displays the expected higher NIS for a positive sample containing susceptible bacteria as compared to a negative and antibiotic-treated sample. This result suggests that our device is highly robust and can be used to determine AST using urine samples with a complex matrix.

Comparison with existing literature

The performance of our device was compared with existing literature, and this is summarized in Supplementary Table S3. This is grouped into: (A) bacterial detection and (B) susceptibility testing. Studies in the first group often use biorecognition elements like lectins or antibodies to correlate impedance changes with bacterial concentration (Supplementary Table S3, Group A). For example, Rengaraj et al. (2018)³⁰ used lectin-immobilized carbon electrodes on paper substrates, achieving a detection limit of 1.9×10^3 CFU/mL. Ruan et al. (2002)³¹ employed antibody-coated indium tin oxide electrodes for detecting 6×10^3 CFU/mL of *E. coli*. Recently, Akhtarian et al. (2024)³² integrated cell-imprinted polymers on stainless steel microwires in a microfluidic device, achieving a detection of 2×10^2 CFU/mL. These methods required ~30 min but were limited to bacterial detection only.

Previous studies on antimicrobial susceptibility testing (Supplementary Table S3, Group B) employed diverse approaches. Swami et al. (2022)²⁷ used low-conductivity zwitterionic buffers with inter-digitated electrodes (IDE) to measure solution resistance changes caused by ionic release from bacteria exposed to surface-acting antibiotics (20 min detection time, 2×10^5 CFU). Growth-based measurements require around 80 min. Hannah et al. (2019)¹⁵ utilized screen-printed gold electrodes (SPE) with nutrient gels containing antibiotics and redox mediators. They correlated impedance changes with bacterial susceptibility (60 min, 5×10^4 CFU). Yang et al., (2020)¹⁷ integrated electrical measurement in nano-constricted microfluidic channels and correlated electrical resistance with susceptibility testing (2 h, 60 CFU). However, these methods rely on costly electrodes (IDE/SPE) or complex microfluidic designs, limiting scalability.

Our device demonstrates high sensitivity, detecting immobilized bacteria at levels as low as ~84 cells/mm² (1764 CFU immobilized). This is equivalent to ~1000 CFU, based on a sample volume of just 10 µL. The detection limit is estimated at 525 immobilized bacteria. This enables accurate detection at the widely accepted diagnostic threshold for urinary tract infections (UTIs) of $> 10^5$ CFU/mL³⁴, with a minimal sample volume. Besides it works directly with the crude urine sample. This is particularly advantageous for rapid, resource-efficient diagnostics, especially in low-resource settings. Furthermore, the device's compact design and reliance on cost-effective screen-printed carbon electrodes ensure affordability and accessibility for widespread clinical use, including high-throughput testing. Compared to conventional methods that often require larger volumes, longer processing times, or expensive instrumentation, our device offers a simplified, faster, and more affordable alternative.

Recent studies propose lowering the diagnostic threshold for UTIs to 10^3 – 10^4 CFU/mL, highlighting the need for even greater sensitivity in detection³⁴. To address this, integrating preconcentration techniques such as acoustic or inertial microfluidics or centrifugation into our existing device is a promising avenue. These advancements would enhance detection capabilities without compromising the device's affordability or simplicity, making it suitable for a broader range of clinical applications and improving diagnostic outcomes in real-world scenarios. This represents a significant future direction for our work.

Conclusions

We propose a lab-on-chip device with a low-cost carbon electrode embedded in a microfluidic device for rapid bacterial detection and susceptibility testing. The diluted low-conductivity medium (10% TNM) was selected as an electrolyte as it has a higher impedance baseline signal and supports the growth of bacteria to obtain high sensitivity of the device (# 84/mm²). Normalised impedance signal (NIS), based on the change in charge transfer resistance, was used as a response signal as an alternative to current widely used solution resistance, particularly in growth-based impedance spectroscopy. The proposed method correlates NIS with the bacterial concentration and susceptibility/resistant profile of bacteria within an incubation time of 3 h. It was demonstrated to be applicable to both gram-positive and gram-negative bacteria and different antibiotic modes of action. In addition, we have demonstrated the device's efficacy in performing AST in human urine samples. With its low-cost, sensitive, and easy-to-use features, the proposed device could enable widespread susceptibility testing to combat antimicrobial resistance, particularly in rural settings.

Materials and methods

Preparation of bacterial culture and antibiotics

Experiments were performed using gram-negative bacteria *Escherichia coli* (ATCC 25922) and gram-positive bacteria *Bacillus subtilis* (ATCC 6051). The cells were grown overnight in a tryptone nutrient medium (TNM) (1.5% w/v, from Himedia Labs, India) at 37 °C, 240 rpm. The culture was centrifuged at 4000 rpm for 5 min, and the pellet was resuspended with 10% TNM. Optimisation steps for obtaining 10% TNM are described in the next section. The optical density of bacterial culture was measured at 600 nm wavelength (A_{600}) using a UV-visible spectrophotometer (Shimadzu UV-1800). It was estimated that a 0.1 optical density (OD) corresponds to 10^8 CFU/mL bacteria using the standard spread plate method. Accordingly, different bacterial concentrations (10^5 – 10^8 CFU/mL) were prepared by serial dilution using 10% TNM. Similar experiments were also performed with the ampicillin-resistant bacteria. The ampicillin-resistant strain was developed by gradually increasing the concentration of ampicillin (10, 30, 50, 70 µg/mL). A standard disk diffusion test confirmed this as per CLSI 2019 guidelines (Supplementary Fig. S8b). To study the effect of nutrient medium, bacteria were also cultured in Luria-Bertani (LB) broth (2.5% w/v, composition of 10 g/L tryptone, 5 g/L NaCl and 5 g/L yeast extract, from Himedia Labs, India). The antibiotics stock solution containing 50 mg/mL of ampicillin sodium salt (SRL, India) and 50 mg/mL of tetracycline (SRL, India) were dissolved in sterile deionised water. All experiments were conducted using ultrapure deionised water (resistivity of ≥ 18.2 MΩ-cm) obtained from the MilliQ IQ7000 unit of Millipore Corporation. The working concentrations of antibiotics were freshly prepared for every set of experiments.

Selection of the electrolyte

The electrolyte was selected based on its ability to provide a higher impedance (electrical signal) to ensure sensitivity and maintain bacterial growth. Different strengths (10, 25, 50, 75 and 100%) of TNM and LB on impedance signal and bacterial growth were studied to optimise electrolyte concentrations. The different strengths of the medium were obtained by diluting using sterile deionised water. For example, 10% TNM was prepared by diluting (1:10) full-strength medium (100% TNM – 1.5% w/v tryptone). The experiments were conducted either in the 10 mL bulk TNM or the LB solution, with a platinum (Pt) electrode of 0.5 mm diameter as the working electrode (WE) and 2 mm as a counter electrode (CE). Conductivity was measured using a conductometer (Hach HQ series Multimeter) for different concentrations of TNM (Supplementary Table S1).

Electrochemical impedance spectroscopy (EIS)

The impedance spectrum was measured using a portable potentiostat (Metrohm DropSens, STAT-I 400s). The potentiostat is connected with the electrochemical microfluidic device using an I-CABSTAT1 cable connector with crocodile clips (2 mm) as shown in Supplementary Fig. S4c. Impedance was measured between the working and counter electrodes with a sinusoidal frequency between 1 MHz to 50 Hz and 50 mV (AC) voltage. The obtained Nyquist plot was fitted with an improved Randle's equivalent circuit model (ECM) using the in-built EIS fitting analyser in DropView 8400 software (Fig. 1c). The model fit provided individual impedance components, which were used to calculate the normalised impedance signal, as explained in the later section (bacterial detection by ϵ - μ D).

Bacterial growth

The effect of different strengths of TNM on bacterial growth was studied using batch experiments. 100 μ L of overnight grown culture was inoculated into 100 mL of TNM of various strengths (10, 25, 50, 75, and 100%). The growth was measured using OD₆₀₀ for every hour till 6 h. The growth curve (Number of CFU/mL Vs time) was obtained using the standard curve (OD Vs CFU/mL). The growth rate, r (slope), was determined by fitting an exponential curve to the log phase of the growth curve. The doubling time was computed as $t_d = \ln(2)/r$ for each nutrient medium.

Also, experiments were conducted to study the effect of bacteria on a 10% TNM impedance signal. For this, various concentrations of bacteria (10^9 , 5×10^8 , and 10^8 CFU/mL) were prepared in 10% TNM, and EIS was measured immediately.

Fabrication of screen-printed electrode and microfluidic device

Electrode fabrication- Carbon electrodes on the glass substrate were fabricated using the screen-printing technique. A pre-determined pattern was created on the polyester screen mesh with photo-curable ink. Carbon graphene paste (C2171023D1, Gwent Sun Chemical) was spread evenly over the screen mask using a plastic palette to deposit electrodes onto the surface of the glass substrate (Blue star slides: 75 $\times$ 25 mm) and incubated at 60 °C for 10 min, followed by 24 h at room temperature (RT). The dimensions of the electrodes were 25 mm (L) $\times$ 0.5 mm (W) with 1 mm spacing between two electrodes. Two sets of electrode pairs, working electrodes (WE) and counter electrodes (CE) (WE1-CE1, WE2-CE2), were used to serve as a replicate for measurement. All four electrodes were functionalised with 30 μ L of 0.1% Poly-L-lysine (PLL) (Sigma Aldrich) and incubated at RT for 2 h to achieve bacterial attachment on the electrode surface.

Microfluidic device fabrication- The polydimethylsiloxane (PDMS) based microfluidic device with the dimensions of 1.5 mm (W) $\times$ 14 mm (L) $\times$ 0.5 mm (D) was employed. The channels were designed using 3D Builder software. The mould was printed using Stratasys J55 prime 3D printer, utilizing PolyJet technology. The required design was printed using VeroUltra WhiteS (proprietary resin of Stratasys) resin. The resin was cured using vat PhotoPolymerization. The designed part was removed from the support base material using the 2.9% alkaline (NaOH) solution. To make the PDMS-based microfluidic device, we mixed the silicon elastomer and curing agent (9:1 ratio w/w) (SYLGARD 184 Silicone Elastomer Kit, Dow Chemical), deposited on the 3D printed mould, and incubated at 60 °C for 12 h. The electrodes and PDMS microfluidic channel were characterized using a surface profiler (Bruker 3D non-contact profiler contour GT). The following parameters were used: White-light interferometry method, 10X magnification, Wavelength: 68.2 nm, scan speed: 4.36 μ m/s. The dimensions were measured using Bruker Vision version 4 software. To determine the length and width of the microfluidic channel, Nikon Ni-E Upright microscope (1X objective lens) was used and measured using ImageJ software. The PDMS device was attached to the PLL functionalised electrode by plasma bond using Harrick Plasma Cleaner (PDC-002). 100 μ L of electrolyte was passed through the channel to remove excess PLL. This device was then used for bacterial detection and susceptibility testing. All devices were used for a single time to avoid contamination.

Bacterial detection by ϵ - μ D

A functionalised microfluidic device is loaded with the bacteria (10^5 – 10^8 CFU/mL), suspended in the electrolyte, and incubated for 30 min to allow bacterial immobilisation. Next, the electrolyte was pumped into the device using a syringe pump (Harvard Apparatus, Pump11 Elite 70–4501) at a 100 μ L/min flow rate for 1 min to support the bacterial growth. This step also removes the sample matrix from the device, which can otherwise potentially affect the electrochemical measurement. A blank experiment (only electrolyte) was also performed without bacteria to obtain the background signal. The proposed detection method relies on the change in impedance, and any evaporation might lead to a background signal; therefore, it is critical to prevent evaporation. This was achieved by saturating the tubes and the tube outlet with the electrolyte (Supplementary Fig. S4b). The EIS spectrum was measured over different incubation times ($t=0$, 3 and 6 h). It was performed under conditions similar to those in the batch experiment. The impedance spectrum was fitted to the improved equivalent circuit

model, and charge transfer resistance (R_{ct}) was obtained each time, as discussed in the result section (optimising electrolyte with bulk solution). The impedance data was then normalised to the corresponding initial signal (R_{ct0}) using the equation below.

Normalised impedance signal (NIS),

$$NIS = \frac{R_{ct} - R_{ct0}}{R_{ct0}} = \frac{(-\Delta R_{ct})}{R_{ct0}}$$

Whereas R_{ct0} is charge transfer resistance at time $t = 0$ h and R_{ct} is charge transfer resistance at different incubation time = t .

Microscopic observation was performed to determine the immobilised bacteria for each initial bacterial concentration. BacLight staining dye (ThermoFisher) was used for staining and observed using an epifluorescence microscope (Nikon Ni-E upright fluorescence microscopy). Staining facilitates the visualisation of the attached bacteria on the electrode. For this, 20 μ L of dye (12 μ M of SYTO-9 and 60 μ M of PI) was passed through the channel using a micropipette and incubated for 15 min to allow complete cell staining. Images were captured at random locations in the device using 50X long working objective lens, and the number of bacteria was counted using ImageJ software with macro batch processing³⁵. The number of bacteria per mm^2 was then calculated using the average number of bacteria in each image and the area of captured images.

Antimicrobial susceptibility test (AST)

Susceptibility testing was conducted by running two parallel devices: a positive control (PC) and an antibiotic susceptibility testing device (AST-device). The protocol for the susceptibility testing is similar to that of bacterial detection. The electrolyte-containing bacteria (10^7 CFU/mL) was loaded into the device and incubated for immobilisation. In the PC, electrolyte without antibiotics was introduced. In the AST-device, electrolyte along with the antibiotic was introduced. For each bacterial strain, various concentrations of antibiotics were tested until a statistically significant change in the NIS of the AST-device was observed with respect to PC. Ampicillin at 10 and 20 μ g/mL concentrations was used for susceptible *E. coli*, whereas 50 and 100 μ g/mL was used for resistant *E. coli*. A low concentration of antibiotic was sufficient to inhibit bacterial growth for susceptible whereas a high antibiotic concentration was required for resistant bacteria. EIS was measured at $t = 0$ h and $t = 3$ h incubation of bacteria in μ D. The NIS for the PC and AST device was compared to determine the susceptibility profile.

A bacterial viability assay was performed using a Live/Dead BacLight kit to confirm the susceptibility profile. Bacteria were stained after incubation with antibiotics in an AST-device using the BacLight stains, as explained in the previous section. SYTO 9 (cell-permeable dye) stains both live and dead cells and emits green fluorescence, whereas propidium iodide (cell-impermeable dye) stains only dead cells and emits red fluorescence. To aid the visualization of fluorescence, the observation is performed at a 100X objective lens. For this, the microfluidic chip is removed from the glass substrate on which the electrode is printed. A cover glass is immediately placed over the substrate and sealed using vacuum grease to prevent evaporation. Microscopic images were captured at 100X objective lens using (a) blue excitation filter (450–490 nm) and barrier filter (515 nm) to capture green fluorescence for visualising live bacteria and (b) green excitation filter (510–560 nm) and barrier filter (575 nm) to capture red fluorescence for visualising dead bacteria.

To show the versatility of our developed AST-device, experiments were performed with other classes of antibiotics and bacteria, such as bacteriostatic antibiotics, tetracycline and gram-positive species, *Bacillus subtilis*. The concentrations of 10 μ g/mL and 5 μ g/mL for ampicillin and tetracycline, respectively, were used for the experiment. Impedance results were confirmed through a standard disk diffusion test following Central Laboratory Standard Institute (CLSI) guidelines²⁹. Images of the experiments were captured using a DSLR camera (Panasonic DMC-GH4) and analysed using ImageJ software to determine the zone of inhibition.

Urine sample experiments

The device was tested and validated using human urine samples. Urine samples were collected from the Institute Hospital at the Indian Institute of Technology Madras (IITM), Chennai, India. The study was approved by the institutional ethics committee of IITM (IEC/2021-03/PS/02). Experiments were performed complying with Indian Council for Medical Research's (ICMR) guidelines. We confirm that informed consent form was obtained from the patients. To test the robustness of the device with the urine sample, the human urine without bacteria (confirmed by pathology report) was directly loaded into the device, and the bacterial detection procedure was carried out. The device was loaded with the electrolyte to remove the urine sample matrix, and the EIS was measured over time to obtain the NIS response. To confirm the bacterial detection in the urine sample, a positive control sample spiked with 10^7 CFU/mL bacteria was used and NIS response was obtained by loading with electrolytes without antibiotics. Finally, the AST protocol was conducted with the positive control sample where electrolyte with tetracycline (5 μ g/mL) was employed.

Statistical analysis

All the experiments were repeated at least in three independent sets for each condition. All results are reported with mean and standard error. All statistical analysis was performed using unpaired student's t-tests with a 95% confidence interval to compare two independent data. For analysis, we used GraphPad Prism 6.01 statistical software.

Data availability

The data set obtained using the present study are available on reasonable request to the corresponding authors.

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Author contributions

R. K. and S. P. secured the funding and administered the project. R. K. and S. G. conceptualized research goals. R. K. and S. P. supervised research activities. Experiments were designed, completed, and analyzed by D. M., S.

G, and R. K. The paper was written by S. G., R. K., D. M, and S. P. S. G., and D. M contributed equally.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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